

Vasamsetti Pavan Manikanta

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Summary

Aspiring **AI Engineer & Data Scientist** skilled in **Machine Learning, Deep Learning, and NLP**. Passionate about **building AI-driven solutions** with expertise in **Python, TensorFlow, Hugging Face, and RAG-based architectures**. Experienced in deploying scalable AI applications, optimizing inference pipelines, and reducing model hallucinations.

Education

Raghu Engineering College Visakhapatnam

B.Tech. in Computer Science and Engineering

Aug 2022 - May 2025

CGPA: 7.56/10

Relevant Coursework: OOPs, Databases, Data Structures, Operating Systems, Computer Networks, Machine Learning.

Skills

Programming & Development: Python, JavaScript, TypeScript, Java, Golang

Machine Learning: Scikit-Learn, XGBoost, CatBoost, Time Series Analysis

Deep Learning: TensorFlow, PyTorch, ResNet, YOLO, LSTMs, GRUs

Natural Language Processing: Hugging Face, Transformers, LangChain, T5, NLTK

Others: Tableau, PostgreSQL, MongoDB, Azure, AWS (EC2, SageMaker, S3), Docker, CI/CD, GitHub, Redis, Linux

Projects

MedQuery AI | [Github](#)

- Developed an **AI-driven Medical QA system** by Fine-tuned **FLAN-T5 with LoRA adapters**, improving medical QA accuracy by **25%**.
- Integrated **Assembly AI** for speech-to-text, reducing manual **input time by 40%**.
- Implemented **RAG (Retrieval-Augmented Generation)** using **Hugging Face's sentence-transformers** and **FAISS vector store**, decreasing hallucinated responses by **60%** through real-time retrieval.
- Built an interactive **UI** for user-friendly medical consultations with real-time response generation.

Skincare Recommendation System | [Github](#)

- Built an **AI skincare assistant** with **YOLO-based acne detection** and **ResNet50 skin classification**.
- Achieved **98% training accuracy, 90% test accuracy** for skin type classification.
- Built an intuitive **Gradio interface**, enabling real-time image analysis and delivering personalized product recommendations.
- Developed an **acne detection** model, identifying blemish types (blackheads, nodules, papules), enhancing **dermatologist assistance**.

Flight Fare Price Prediction | [Github](#)

- Developed a **GRU-based deep learning model** for predicting domestic flight fares, integrating **cyclic time encoding** to capture flight scheduling patterns.
- Engineered domain-specific features** (seasonality, holidays, weekends) to improve prediction accuracy.
- Trained the model with early stopping** over 800 epochs, achieving **R²: 0.8852, RMSE: 1558.42** for precise fare estimation.
- Optimized data preprocessing using **RobustScaler & OneHotEncoder**, ensuring performance stability in noisy datasets.

Certificates

- Generative AI with Large Language Models** – Coursera
- Supervised Machine Learning** – Coursera
- Go Beyond the Numbers (Data Analytics)** – Coursera
- Backend Development and APIs** – FreeCodeCamp